String charAt() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringCharAtDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String charAt() Method using the exact Java program maintained in Assignment 3.

System Requirements

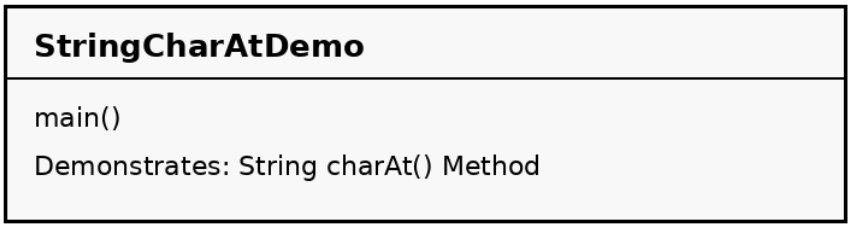
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String charAt() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringCharAtDemo {
public static void main(String[] args) {

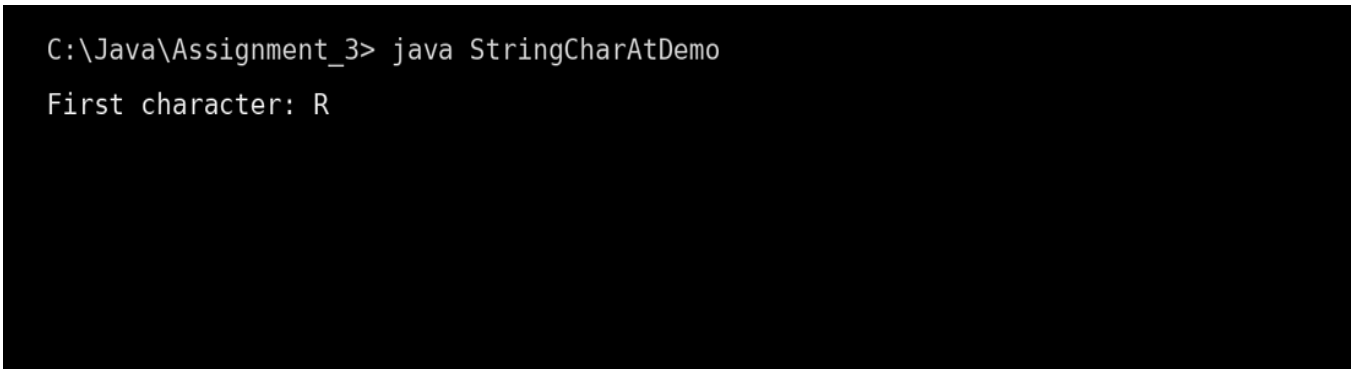
String username = "Rahul";

// Get the first character
System.out.println("First character: " + username.charAt(0));
}
}
```

Compilation & Execution Commands

Save as: StringCharAtDemo.java
Compile: javac StringCharAtDemo.java
Execute: java StringCharAtDemo

Output Screen



Result

Thus, the string charat() method was implemented and executed successfully.

String concat() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringConcatDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String concat() Method using the exact Java program maintained in Assignment 3.

System Requirements

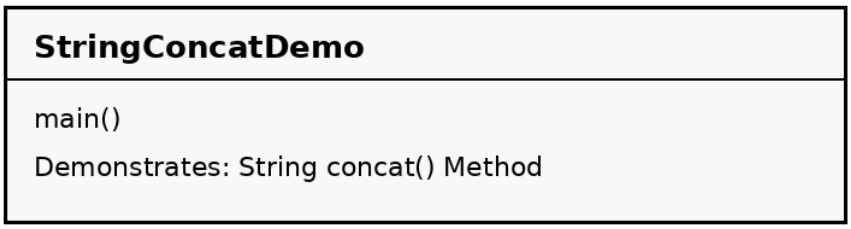
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String concat() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringConcatDemo {
public static void main(String[] args) {

String firstName = "Rahul";
String lastName = "Kumar";

// Join two strings
String fullName = firstName.concat(" ").concat(lastName);

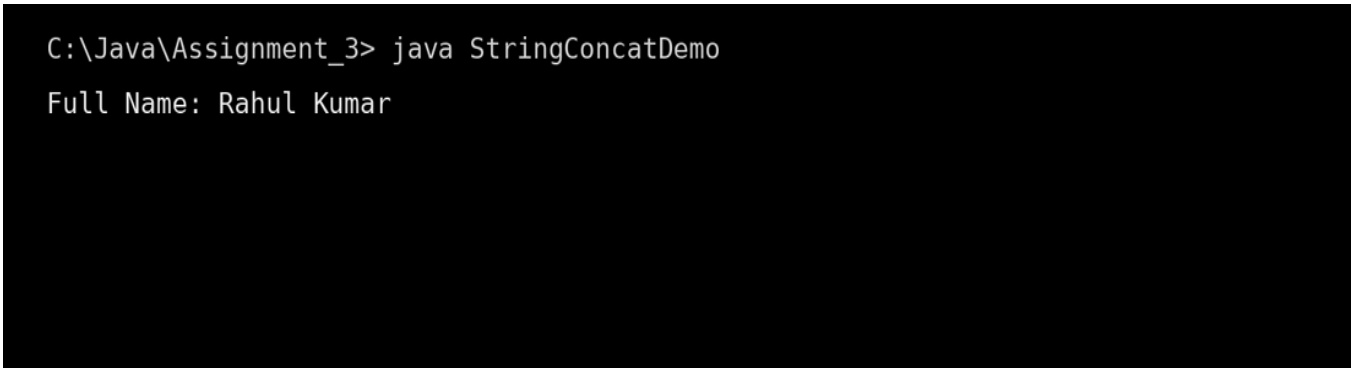
System.out.println("Full Name: " + fullName);
}
```

```
}
```

Compilation & Execution Commands

Save as: StringConcatDemo.java
Compile: javac StringConcatDemo.java
Execute: java StringConcatDemo

Output Screen



Result

Thus, the string concat() method was implemented and executed successfully.

String contains() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringContainsDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String contains() Method using the exact Java program maintained in Assignment 3.

System Requirements

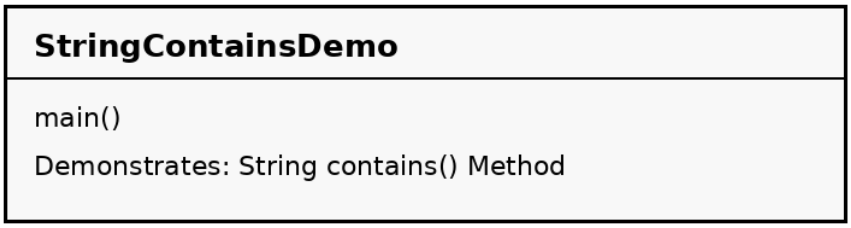
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String contains() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringContainsDemo {
public static void main(String[] args) {

String email = "rahul@gmail.com";

// Check whether email contains Gmail
if (email.contains("gmail")) {
System.out.println("Gmail account");
}
}
}
```

Compilation & Execution Commands

Save as: StringContainsDemo.java
Compile: javac StringContainsDemo.java
Execute: java StringContainsDemo

Output Screen

```
C:\Java\Assignment_3> java StringContainsDemo
Gmail account
```

Result

Thus, the string contains() method was implemented and executed successfully.

String endsWith() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringEndsWithDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String endsWith() Method using the exact Java program maintained in Assignment 3.

System Requirements

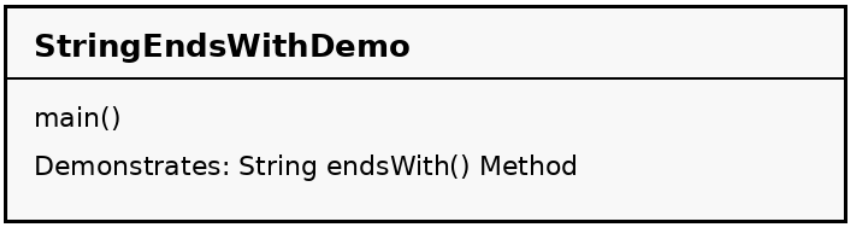
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String endsWith() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringEndsWithDemo {
public static void main(String[] args) {

String fileName = "report.pdf";

// Check the file type
if (fileName.endsWith(".pdf")) {
System.out.println("PDF file");
}
}
}
```

Compilation & Execution Commands

Save as: StringEndsWithDemo.java
Compile: javac StringEndsWithDemo.java
Execute: java StringEndsWithDemo

Output Screen

```
C:\Java\Assignment_3> java StringEndsWithDemo
PDF file
```

Result

Thus, the string endswith() method was implemented and executed successfully.

String equals() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringEqualsDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String equals() Method using the exact Java program maintained in Assignment 3.

System Requirements

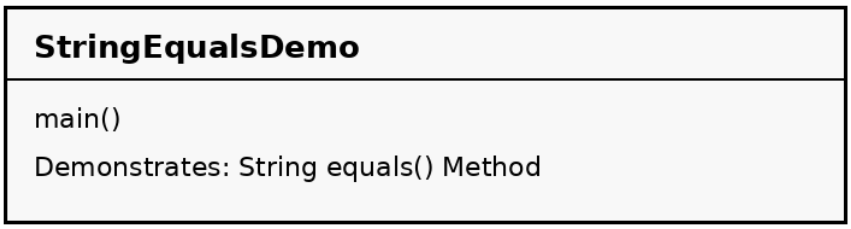
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String equals() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringEqualsDemo {
public static void main(String[] args) {

String enteredPassword = "java123";
String correctPassword = "java123";

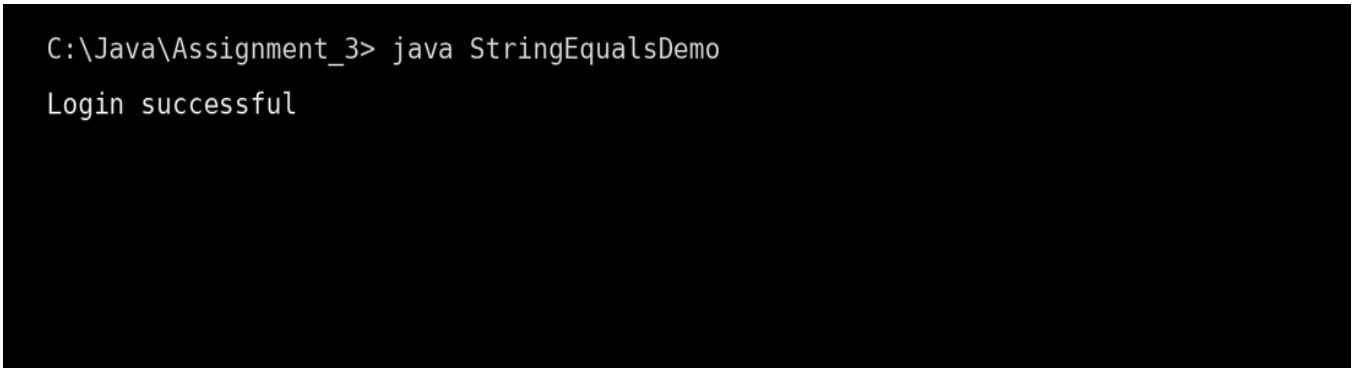
// Compare the actual string values
if (enteredPassword.equals(correctPassword)) {
System.out.println("Login successful");
} else {
System.out.println("Wrong password");
```

```
}
}
}
```

Compilation & Execution Commands

Save as: StringEqualsDemo.java
Compile: javac StringEqualsDemo.java
Execute: java StringEqualsDemo

Output Screen



Result

Thus, the string equals() method was implemented and executed successfully.

String equalsIgnoreCase() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringEqualsIgnoreCaseDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String equalsIgnoreCase() Method using the exact Java program maintained in Assignment 3.

System Requirements

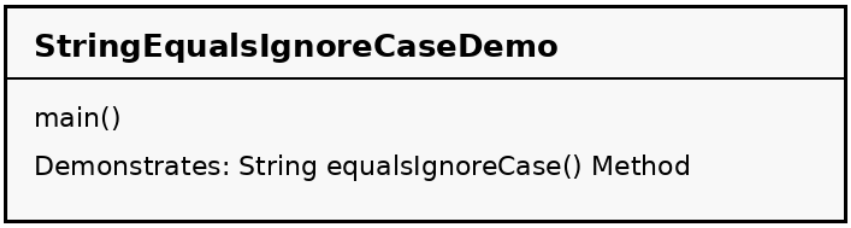
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String equalsIgnoreCase() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringEqualsIgnoreCaseDemo {
public static void main(String[] args) {

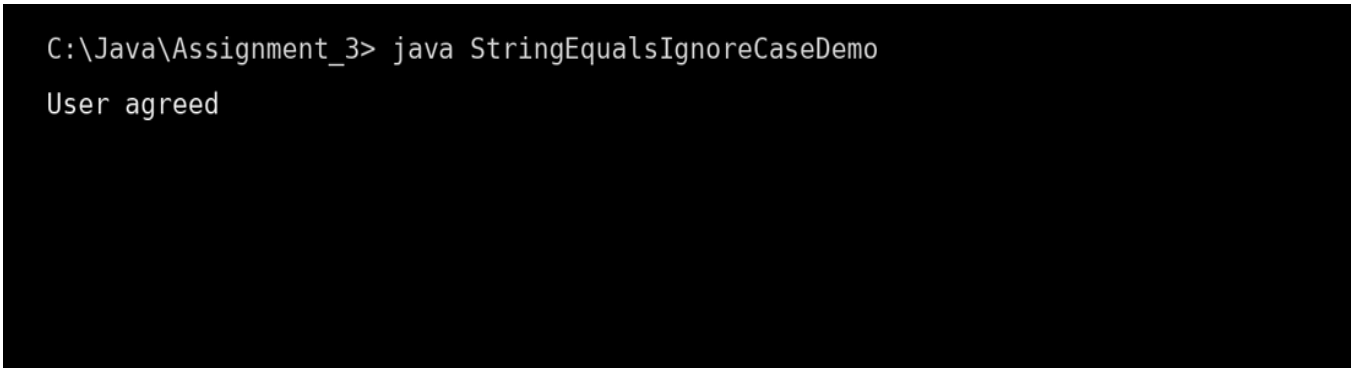
String answer = "YES";

// Ignore uppercase and lowercase differences
if (answer.equalsIgnoreCase("yes")) {
System.out.println("User agreed");
}
}
}
```

Compilation & Execution Commands

Save as: StringEqualsIgnoreCaseDemo.java
Compile: javac StringEqualsIgnoreCaseDemo.java
Execute: java StringEqualsIgnoreCaseDemo

Output Screen



Result

Thus, the string equalsignorecase() method was implemented and executed successfully.

String indexOf() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringIndexOfDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String indexOf() Method using the exact Java program maintained in Assignment 3.

System Requirements

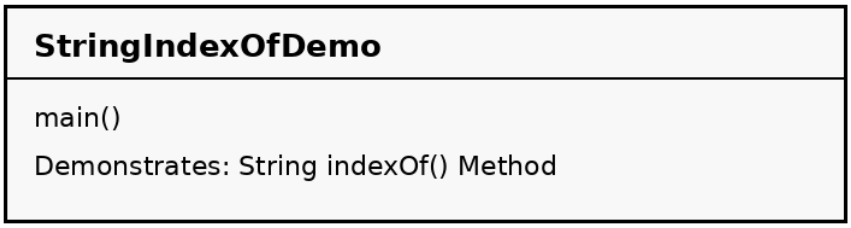
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String indexOf() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringIndexOfDemo {
public static void main(String[] args) {

String email = "rahul@gmail.com";

// Find the position of @
int position = email.indexOf("@");

System.out.println("@ position: " + position);
}
}
```

Compilation & Execution Commands

Save as: StringIndexOfDemo.java
Compile: javac StringIndexOfDemo.java
Execute: java StringIndexOfDemo

Output Screen

```
C:\Java\Assignment_3> java StringIndexOfDemo
@ position: 5
```

Result

Thus, the string indexof() method was implemented and executed successfully.

String length() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringLengthDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String length() Method using the exact Java program maintained in Assignment 3.

System Requirements

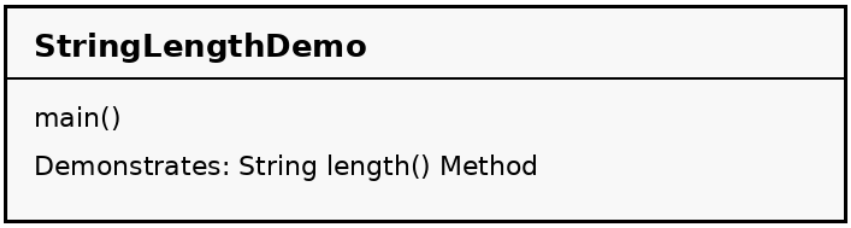
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String length() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringLengthDemo {
public static void main(String[] args) {

String username = "rahul123";

// Check username length
System.out.println("Length: " + username.length());
}
}
```

Compilation & Execution Commands

Save as: StringLengthDemo.java
Compile: javac StringLengthDemo.java
Execute: java StringLengthDemo

Output Screen



Result

Thus, the string length() method was implemented and executed successfully.

String toLowerCase() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringLowerCaseDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String toLowerCase() Method using the exact Java program maintained in Assignment 3.

System Requirements

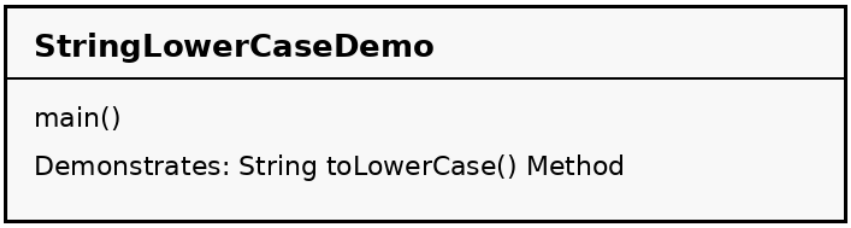
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String toLowerCase() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringLowerCaseDemo {
public static void main(String[] args) {

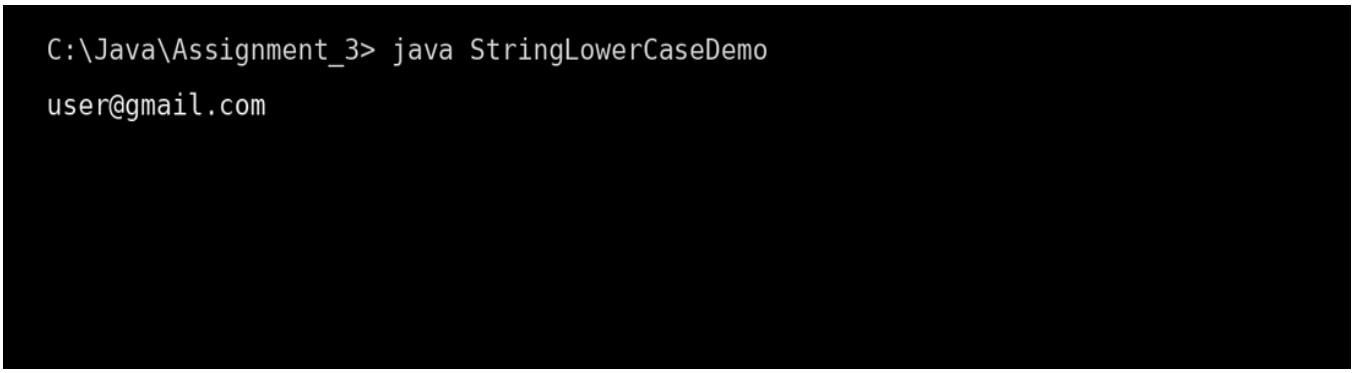
String email = "USER@GMAIL.COM";

// Convert email to lowercase
System.out.println(email.toLowerCase());
}
}
```

Compilation & Execution Commands

Save as: StringLowerCaseDemo.java
Compile: javac StringLowerCaseDemo.java
Execute: java StringLowerCaseDemo

Output Screen



Result

Thus, the string tolowercase() method was implemented and executed successfully.

String replace() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringReplaceDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String replace() Method using the exact Java program maintained in Assignment 3.

System Requirements

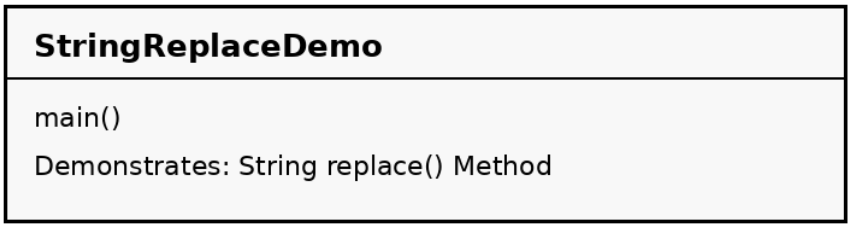
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String replace() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringReplaceDemo {
public static void main(String[] args) {

String message = "Welcome Java";

// Replace Java with Python
String newMessage = message.replace("Java", "Python");

System.out.println(newMessage);
}
}
```

Compilation & Execution Commands

Save as: StringReplaceDemo.java
Compile: javac StringReplaceDemo.java
Execute: java StringReplaceDemo

Output Screen



Result

Thus, the string replace() method was implemented and executed successfully.

String split() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringSplitDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String split() Method using the exact Java program maintained in Assignment 3.

System Requirements

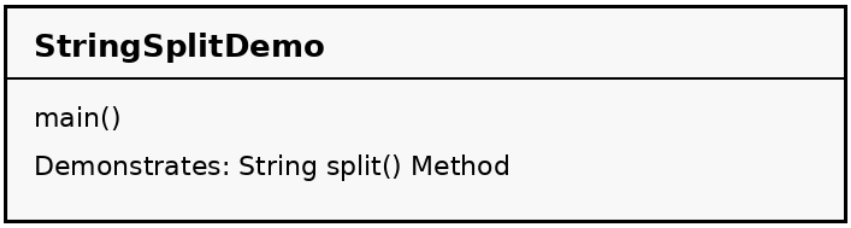
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String split() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringSplitDemo {
public static void main(String[] args) {

String skills = "Java,Python,SQL";

// Split the string using comma
String[] skillList = skills.split(",");

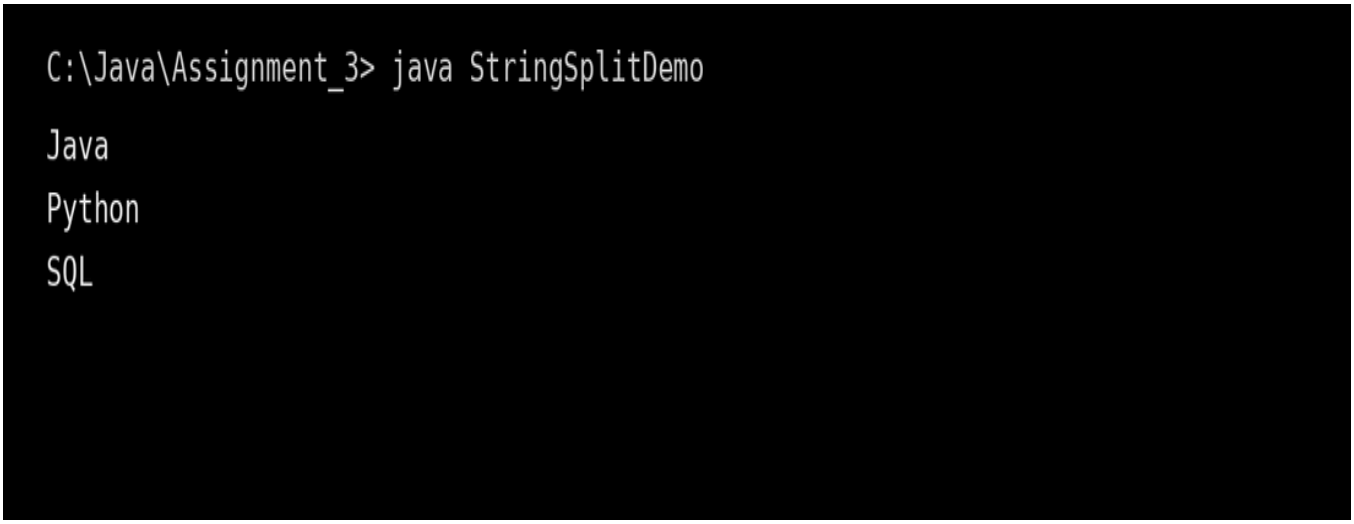
for (String skill : skillList) {
System.out.println(skill);
}
```

```
}
}
```

Compilation & Execution Commands

Save as: StringSplitDemo.java
Compile: javac StringSplitDemo.java
Execute: java StringSplitDemo

Output Screen



Result

Thus, the string split() method was implemented and executed successfully.

String startsWith() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringStartsWithDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String startsWith() Method using the exact Java program maintained in Assignment 3.

System Requirements

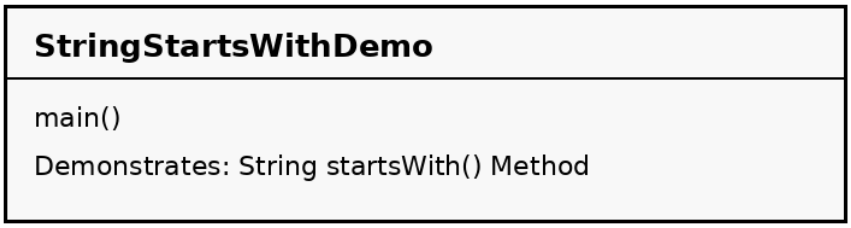
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String startsWith() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringStartsWithDemo {
public static void main(String[] args) {

String phone = "+91 9876543210";

// Check the country code
if (phone.startsWith("+91")) {
System.out.println("Indian number");
}
}
}
```

Compilation & Execution Commands

Save as: StringStartsWithDemo.java
Compile: javac StringStartsWithDemo.java
Execute: java StringStartsWithDemo

Output Screen

```
C:\Java\Assignment_3> java StringStartsWithDemo
Indian number
```

Result

Thus, the string startswith() method was implemented and executed successfully.

String substring() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringSubstringDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String substring() Method using the exact Java program maintained in Assignment 3.

System Requirements

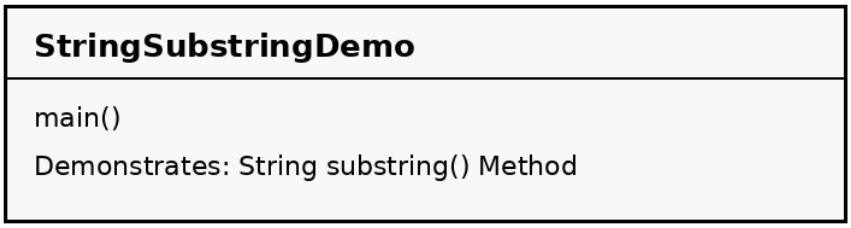
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String substring() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringSubstringDemo {
public static void main(String[] args) {

String email = "rahul@gmail.com";

// Extract username before @
String username = email.substring(0, email.indexOf("@"));

System.out.println("Username: " + username);
}
}
```

Compilation & Execution Commands

Save as: StringSubstringDemo.java
Compile: javac StringSubstringDemo.java
Execute: java StringSubstringDemo

Output Screen

```
C:\Java\Assignment_3> java StringSubstringDemo
Username: rahul
```

Result

Thus, the string substring() method was implemented and executed successfully.

String trim() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringTrimDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String trim() Method using the exact Java program maintained in Assignment 3.

System Requirements

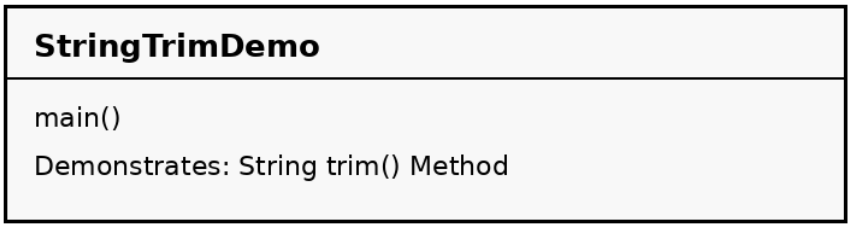
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String trim() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringTrimDemo {
public static void main(String[] args) {

String name = " Rahul ";

// Remove spaces from both ends
System.out.println("Name: " + name.trim());
}
}
```

Compilation & Execution Commands

Save as: StringTrimDemo.java
Compile: javac StringTrimDemo.java
Execute: java StringTrimDemo

Output Screen



Result

Thus, the string trim() method was implemented and executed successfully.

String toUpperCase() Method in Java

Basic Information

Assignment: Assignment 3

Program: StringUpperCaseDemo.java

Topic: String Methods

Objective / Aim

To demonstrate the String toUpperCase() Method using the exact Java program maintained in Assignment 3.

System Requirements

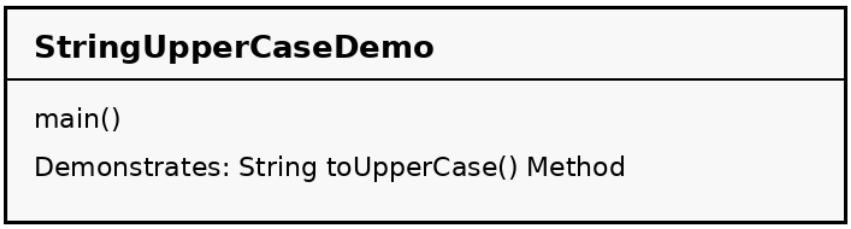
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the String toUpperCase() method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



Source Code

```
public class StringUpperCaseDemo {
public static void main(String[] args) {

String city = "visakhapatnam";

// Convert city name to uppercase
System.out.println(city.toUpperCase());
}
}
```

Compilation & Execution Commands

Save as: StringUpperCaseDemo.java
Compile: javac StringUpperCaseDemo.java
Execute: java StringUpperCaseDemo

Output Screen



Result

Thus, the string touppercase() method was implemented and executed successfully.

Case Conversion using String Methods in Java

Basic Information

Assignment: Assignment 3

Program: case_convertion.java

Topic: String Methods

Objective / Aim

To demonstrate the Case Conversion using String Methods using the exact Java program maintained in Assignment 3.

System Requirements

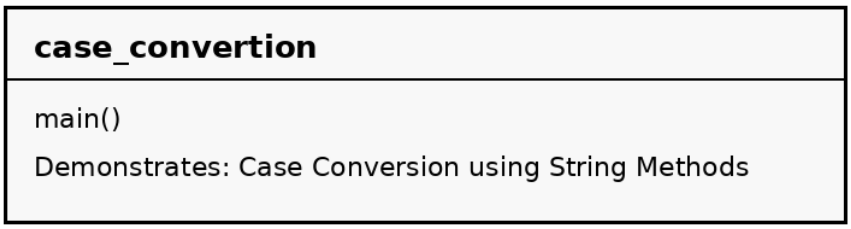
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create the required String value.
3. Apply the Case Conversion using String method as shown in the source code.
4. Display the result of the String operation.
5. Stop the program.

UML Class Diagram



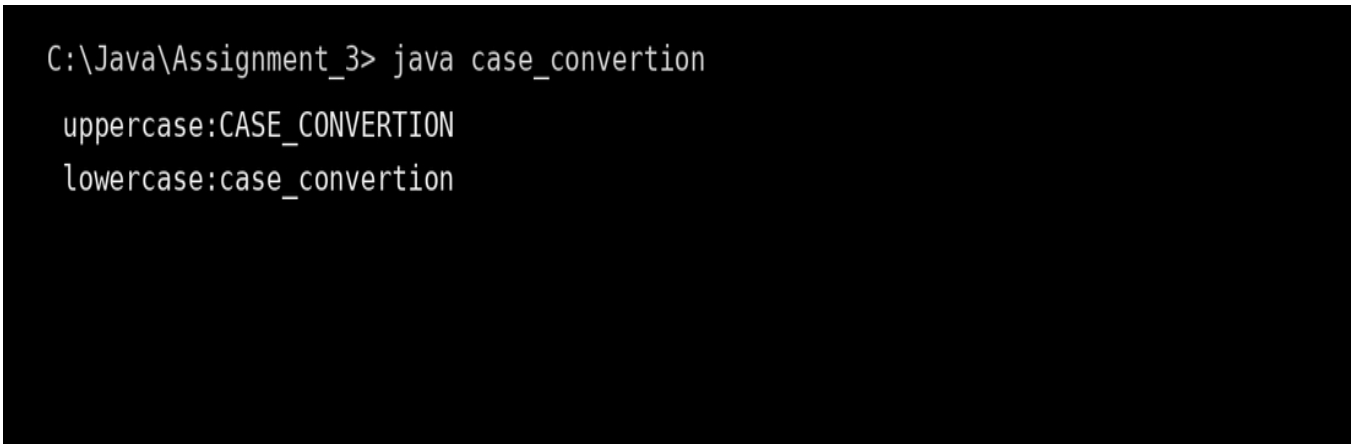
Source Code

```
public class case_convertion {
public static void main(String[] args) {
String name= "Case_convertion";
System.out.println(" uppercase:"+name.toUpperCase());
System.out.println(" lowercase:"+name.toLowerCase());
}
}
```

Compilation & Execution Commands

Save as: case_convertion.java
Compile: javac case_convertion.java
Execute: java case_convertion

Output Screen



Result

Thus, the case conversion using string methods was implemented and executed successfully.

String indexOf() Example in Java

Basic Information

Assignment: Assignment 3

Program: indexExample.java

Topic: String Methods

Objective / Aim

To demonstrate the String indexOf() Example using the exact Java program maintained in Assignment 3.

System Requirements

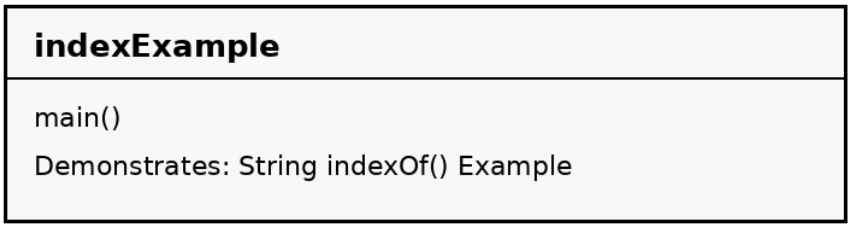
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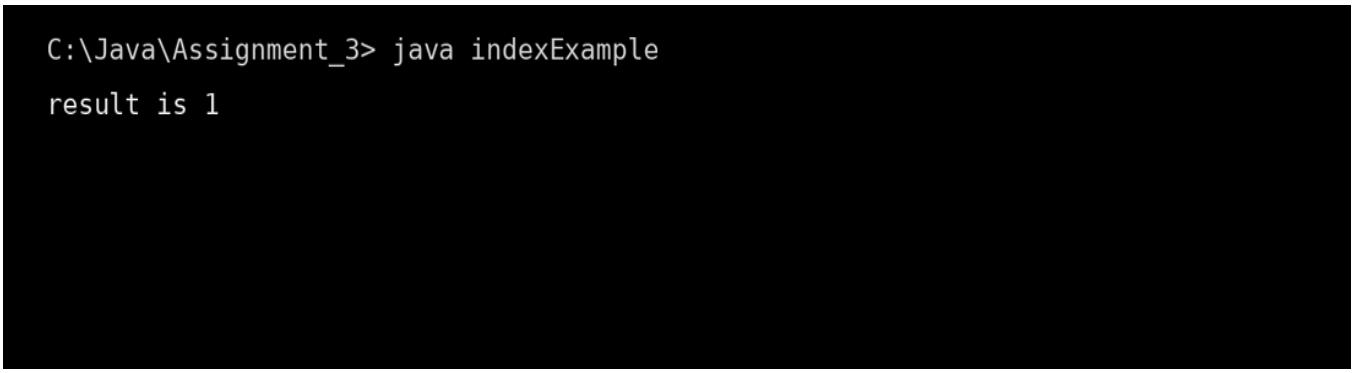
Source Code

```
public class indexExample{
public static void main(String[] args) {
String name= "venky";
int result =name.indexOf("e");
System.out.println("result is "+ result);
}
}
```

Compilation & Execution Commands

Save as: indexExample.java
Compile: javac indexExample.java
Execute: java indexExample

Output Screen



Result

Thus, the string indexof() example was implemented and executed successfully.